

Moving-CLAP: Spatially Aware Audio–Language Embeddings for Moving Sound Sources

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Abstract—Contrastive language–audio pretraining (CLAP) provides a foundation for audio–language embeddings. Although several spatially-aware extensions have been developed, they are limited to static sources, leaving the modeling of moving sound sources as a critical challenge. We thus propose *Moving-CLAP*, a framework designed to capture time-varying sound source directions. We integrate an audio encoder recognizing the direction of a sound source in frame-level segments with a text encoder enhanced by *spatially-biased attention* that emphasizes directional tokens. We introduce two loss functions to improve spatial discriminability and directional consistency of temporal changes. Experiments show that Moving-CLAP learns embeddings that capture single-source motion, particularly its start and end directions, while effectively structuring the embedding space by spatial information. The embeddings also outperform the conventional framework in capturing two-source motions.

Index Terms—Audio–language embeddings, moving sound sources, CLAP, audio–language model

1. INTRODUCTION

Contrastive language–audio pretraining (CLAP)-based models [1]–[3] have been proposed to project audio signals and natural language descriptions into a shared embedding space. These models are especially effective at capturing *content* information—representing what sound events are occurring—in audio signals and have been widely used as encoders for various downstream tasks, such as audio–text retrieval [1], [4], [5], automated audio captioning [6]–[8], and text-to-audio generation [9]–[11]. However, these models focus on monaural audio signals, which prevents models from capturing spatial information such as sound source direction.

To recognize spatial information, spatially-aware CLAP-based models that integrate spatial information with content information have been proposed. Frameworks such as ELSA [12] and Spatial-CLAP [13] perform contrastive learning by introducing a specialized spatial encoder in addition to the conventional audio encoder, which functions as a content encoder in these models. The content encoder extracts content information (e.g., sound events), while the spatial encoder extracts *spatial* information—representing where these events are located, typically via the direction of arrival (DoA). In contrast to sound event localization and detection (SELD) models [14]–[17], which are restricted to predefined classes and rigid coordinate outputs, these language-aligned models offer a flexible, open-vocabulary approach to spatial audio understanding.

However, existing spatially-aware CLAP-based models are limited to static sources. Unlike SELD models, which can track source motions, these models typically estimate only a single DoA from an entire audio signal for each source. Consequently, they fail to account for moving sources and to establish a correspondence between content information and source motion [18].

To address these limitations, we propose *Moving-CLAP*, a framework to learn audio–language embeddings that treats content information and time-varying DoAs for all sources in a unified manner. As

Table 1: Comparison of framework capabilities.

Framework	Semantic description	Spatial awareness	Multi-source	Moving-source
SELD [14]–[17]	Limited	✓	✓	✓
CLAP [1]–[3]	✓			
ELSA [12]	✓	✓		
Spatial-CLAP [13]	✓	✓	✓	
Ours	✓	✓	✓	✓

shown in Table 1, our framework enables the learning of embeddings for moving sound sources. The contributions of our study are as follows¹:

- **Model architecture:** We integrate an audio encoder derived from SELD with a text encoder enhanced by *spatially-biased attention*, significantly improving the extraction of temporal features from moving sound sources and their captions.
- **Training strategies:** We introduce *swap* and *spatial contrastive losses* to improve spatial discriminability and the directional consistency of temporal changes in spatial information, enhancing the separation and structural organization of content and spatial information.
- **Moving source evaluation:** We establish a training and evaluation framework using a created moving sound source dataset. We show that our framework learns embeddings achieving high retrieval performance for source motion and structures the embedding space based on spatial information.

2. PROPOSED FRAMEWORK: MOVING-CLAP

2.1. Model architecture

Fig. 1 shows the overview of Moving-CLAP. This framework consists of an audio encoder and a text encoder. The audio encoder adopts a parallel architecture in which a stereo input is split into two separate paths: a spatial encoder and a content encoder. The outputs from these encoders are concatenated through feature fusion and temporal pooling.

Spatial encoder: The spatial encoder captures spatial information consisting of time-varying DoAs. We adopt an architecture based on SELDnet [14], capable of multi-source detection and localization. Unlike Spatial-CLAP which applies global pooling in its spatial encoder, we use the frame-wise output of the spatial encoder, thereby preserving the time-varying characteristics of spatial information. We obtain D_{sp} -dimensional temporal-spatial features $\mathbf{F}_{sp} \in \mathbb{R}^{T \times D_{sp}}$, where T denotes the number of time frames.

Content encoder: The content encoder captures content information, representing what sound events are occurring in the audio signal. The stereo input is averaged into a monaural signal before encoding. Instead of HTS-AT [19] used in Spatial-CLAP, we utilize BEATs [20] to leverage pre-trained weights with high temporal resolution, as detailed in Section 3.2. The D_{src} -dimensional content features extracted by

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¹<https://github.com/sarulab-speech/Moving-CLAP>

BEATs are resampled via linear interpolation to align their temporal resolution with $\mathbf{F}_{\text{sp}}$, yielding $\mathbf{F}_{\text{src}} \in \mathbb{R}^{T \times D_{\text{src}}}$.

Feature fusion and pooling: The features obtained from the content and spatial encoders are integrated using a multihead attention block (MAB) [21]. To compress the temporal information, the fused features are aggregated along the time axis via an attention-based pooling layer using a learnable query. Finally, a multilayer perceptron (MLP) and L_2 normalization are applied to obtain the final audio embedding $\mathbf{z}^a$.

Text encoder: We employ a pre-trained RoBERTa [22] model. To capture spatial cues in captions, we enhance the text encoder with *spatially-biased attention* (SBA), which is an attention mechanism designed to focus on predefined spatial descriptors by assigning them higher importance scores. Specifically, an MLP calculates base importance scores for each token, and a positive bias is then added to the scores of the predefined spatial descriptors and [CLS] token [23]. This design prioritizes directional tokens while preserving the underlying content information. The final text embedding $\mathbf{z}^t$ is obtained by computing a weighted average of the token sequence, where weights are derived by applying a softmax function to the biased importance scores, followed by an MLP and L_2 normalization.

2.2. Training strategy

The training of Moving-CLAP is conducted in two stages: (i) pre-training of the spatial encoder on the SELD task for moving sound sources, and (ii) joint optimization of the entire model to align audio and text embeddings.

2.2.1. Pre-training of the spatial encoder: To enable the spatial encoder to track moving sources, we pre-train it on the SELD task using the SELDnet framework. Specifically, the spatial features $\mathbf{F}_{\text{sp}}$ and content features $\mathbf{F}_{\text{src}}$ are input to two task-specific decoders. One decoder for the sound event detection (SED) task estimates frame-level event probabilities using $\mathbf{F}_{\text{src}}$. The other decoder for the DoA estimation task estimates frame-level DoAs for each sound event class by concatenating both $\mathbf{F}_{\text{sp}}$ and $\mathbf{F}_{\text{src}}$. Since this study focuses on horizontal motion, the output of the DoA decoder is represented as a two-dimensional vector $(\cos \theta, \sin \theta)$ normalized by a hyperbolic tangent function. We employ the binary cross-entropy loss L_{SED} for SED and a cosine distance loss L_{DoA} for DoA estimation. The total pre-training loss function is defined as $L_{\text{SELD}} = L_{\text{SED}} + \lambda_{\text{DoA}} L_{\text{DoA}}$ where λ_{DoA} is a weighting hyperparameter.

2.2.2. Joint optimization of the entire model: Moving-CLAP is trained through joint optimization of the CLIP loss L_{CLIP} [1] and two novel auxiliary losses: *swap loss* and *spatial contrastive loss*. While L_{CLIP} aligns audio–text pairs, it struggles to decouple spatial information from content information. In random batches, which rarely contain identical sound events with diverse source motion, the model tends to rely on content information alone for retrieval, failing to learn discriminative spatial embeddings. Inspired by ELISA, which incorporates auxiliary regression tasks (e.g., DoA or distance estimations) to strengthen spatial awareness, we propose two auxiliary losses.

Swap loss L_{swap} : To enhance directional discriminability, this loss employs direction-swapped captions as hard negatives. For a set of moving audio signals $\mathcal{M}$, let $\mathbf{z}_{i,\text{sw}}^t$ be the swapped text embedding generated by swapping the directional descriptors of audio signal $i \in \mathcal{M}$ (e.g., changing “left side to front” to “front to left side”). The swap loss is defined as

$$L_{\text{swap}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \sum_{k \in \{\text{TT}, \text{AT}\}} \max(0, m_k - \Delta_k), \quad (1)$$

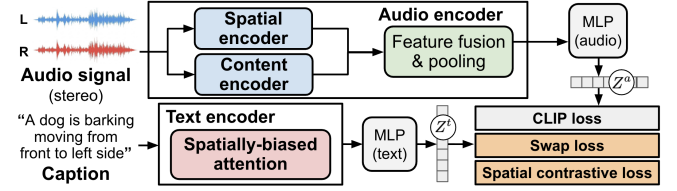


Fig. 1: Overview of Moving-CLAP

where $\Delta_{\text{TT}} = 1 - f_{\text{sim}}(\mathbf{z}_i^t, \mathbf{z}_{i,\text{sw}}^t)$ and $\Delta_{\text{AT}} = f_{\text{sim}}(\mathbf{z}_i^a, \mathbf{z}_i^t) - f_{\text{sim}}(\mathbf{z}_i^a, \mathbf{z}_{i,\text{sw}}^t)$. Here, $f_{\text{sim}}(\cdot, \cdot)$ denotes cosine similarity. The former, Δ_{TT} , encourages separation within the text modality by enforcing the embedding distance to exceed a margin m_{TT} , aiming to map linguistically similar but spatially opposite descriptions to distant embeddings. Simultaneously, Δ_{AT} enforces cross-modal alignment by requiring the audio embedding $\mathbf{z}_i^a$ to be closer to the correct caption than to the swapped one by a margin m_{AT} .

Spatial contrastive loss L_{spatial} : This loss facilitates the discrimination of spatial patterns from a predefined set of K candidate motions $\mathcal{C} = \{c_k\}_{k=1}^K$. Each spatial pattern is defined as a pair of start and end directions between predefined spatial regions (e.g., “front-left to front”). For a set of spatial audio signals $\mathcal{S}$, let $\mathbf{z}_k^t$ be the text embedding for candidate pattern c_k , and $y_i \in \{1, \dots, K\}$ be the index of the ground-truth pattern for a given audio signal i . Based on the InfoNCE loss [24], the spatial contrastive loss is defined as

$$L_{\text{spatial}} = -\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \log \frac{\exp(f_{\text{sim}}(\mathbf{z}_i^a, \mathbf{z}_{y_i}^t)/\tau_{\text{sp}})}{\sum_{k=1}^K \exp(f_{\text{sim}}(\mathbf{z}_i^a, \mathbf{z}_k^t)/\tau_{\text{sp}})}, \quad (2)$$

where τ_{sp} is a temperature parameter. By treating these spatial patterns as distinct classes, this loss function stabilizes the recognition of directional changes independently of the specific batch composition.

The total loss L_{total} is defined as $L_{\text{total}} = L_{\text{CLIP}} + \lambda_{\text{sw}} L_{\text{swap}} + \lambda_{\text{sp}} L_{\text{spatial}}$, where λ_{sw} and λ_{sp} are hyperparameters.

3. EXPERIMENT

3.1. Dataset

We extended AudioCaps [25] by spatializing its audio signals and captions to include moving sound sources¹.

Simulation of spatial audio: Monaural signals from AudioCaps were spatialized through convolution with room impulse responses (RIRs) generated via the image method [26] using gpuRIR [27]. We adopted a time-variant convolution approach inspired by the DCASE 2020 Task 3 dataset [28].

We considered nine distinct acoustic configurations by combining three room sizes ($6.0 \times 5.0 \times 3.0$, $8.0 \times 6.0 \times 3.5$, and $10.0 \times 8.0 \times 4.0$ m) with three absorption coefficients (0.3, 0.4, 0.5). A stereo microphone array with an inter-element spacing of 0.17 m was placed at the center of each room at a height of 1.5 m. All audio signals were sampled at 16 kHz.

In order to simulate moving sources, we designed movement trajectories of the sound sources. Since stereo microphones inherently have the front-back ambiguity, we restricted the horizontal range for trajectory generation to the frontal semicircle (-90° to 90°). This semicircle was divided into five spatial segments: left side, front-left, front, front-right, and right side. Within these segments, linear and circular trajectories were designed under the following constraints: a minimum distance of 0.5 m from the walls and 1.0 m (linear case) or one-fifth of the room depth (circular case) from the microphone array. To avoid ambiguous categorization, the start and end points of each trajectory were required to be at least 8° away from segment boundaries.

RIRs between the stereo microphone and 100 uniformly discretized points along the source trajectory were generated. Monaural signals were segmented into overlapped frames with half-overlap. Each frame was convolved with the RIRs corresponding to the trajectory point closest to the source position at the center time of the frame. To avoid artifacts caused by discrete RIR switching, the output signals were reconstructed using an overlap-add technique [29] using a triangular window.

Table 2 shows the number of samples in the created dataset. To ensure experimental independence, we generated distinct sets of RIRs for the training, validation, and test subsets by using different source trajectories. These subsets consist of static sources (DoAs change $< 5^\circ$ and within a single spatial segment) and moving sources (DoAs change $\geq 20^\circ$ and transitioning between spatial segments). We generated 25 single-source patterns, consisting of 5 static and 20 moving scenarios from all possible start-to-end spatial segment combinations. For two-source scenarios, we created 180 patterns, including 20 where both sources are static and 160 with one static and one moving source, excluding pairs with the same start positions or two moving sources. To ensure clarity in evaluation, each pair was composed of distinct audio events.

Spatial caption construction: We extended the original captions with spatial descriptors based on the trajectories of the simulated sources. The following templates were employed:

- **Static source:** $\langle \text{event} \rangle \langle \text{location} \rangle$.
- **Moving source:** $\langle \text{event} \rangle$ moving from $\langle \text{start} \rangle$ to $\langle \text{end} \rangle$.

The $\langle \text{location} \rangle$ attribute was selected from [“on the left side”, “in the front-left”, “in front”, “in the front-right”, “on the right side”], while $\langle \text{start} \rangle$ and $\langle \text{end} \rangle$ were assigned from the corresponding directional terms. For two-source scenarios, the individual captions were concatenated with “and”. We also defined “spatial-only” captions by omitting the event content to specifically facilitate the learning of spatial discriminative capabilities.

3.2. Training setting

The spatial encoder was pre-trained for 50 epochs using the Adam optimizer [30], with a batch size of 32 and $\lambda_{\text{DoA}} = 10$. Throughout both pre-training and joint optimization, the output dimensions for the spatial and content encoders were maintained at $D_{\text{sp}} = 256$ and $D_{\text{src}} = 768$, respectively.

To capture the temporal features of moving sound sources, the content encoder was initialized with weights pre-trained on the strongly labeled AudioSet [31], as provided in [32]. We used the RoBERTa model fine-tuned with Spatial-CLAP [33].

The overall framework was first trained on AudioCaps for 20 epochs, then fine-tuned on our moving-source dataset for 40 epochs using the AdamW optimizer [34] with a plateau-decay scheduler and automatic mixed precision [35]. Each batch had a size of 32 and was constructed via balanced sampling of static, moving, and two-source patterns, with no duplicate sound events within a batch. During the joint optimization of the entire model, RoBERTa and the spatial encoder were frozen, while only the final two layers of the content encoder were unfrozen for stabilizing the training.

The final shared embedding was 512-dimensional, and the MAB used 8 heads. For the SBA, spatial descriptors corresponded to the five spatial segments defined in the dataset. For the auxiliary losses, weighting factors and margins were tuned through preliminary experiments, resulting in $\lambda_{\text{sw}} = 0.3$, $\lambda_{\text{sp}} = 0.01$, $m_{\text{TT}} = 0.3$, and $m_{\text{AT}} = 0.2$. A learnable temperature parameter τ_{sp} , constrained within $[0.01, 100.0]$, was assigned to scale the logits. The candidate set $\mathcal{C}$ for the spatial contrastive loss consists of 25 single-source and 90

Table 2: Composition of spatial patterns and number of samples. The top header distinguishes between the original monaural base audio (Mono) and the expanded stereo samples (Stereo) generated across spatial categories.

Subset	Mono	Stereo				
	Base audio	1-src		2-src		Total
		Sta.	Mov.	Sta.+Sta.	Sta.+Mov.	
Train	49,838	24,919	49,838	24,918	49,835	149,510
Validation	2,475	1,237	2,475	1,237	2,472	7,421
Test (single)	975	4,875	19,500	-	-	24,375
Test (two)	(200)	-	-	4,000	32,000	36,000

two-source spatial-only captions, with the latter reduced from 180 patterns due to source-order invariance.

3.3. Compared frameworks

We compared the following two frameworks, including their variants and ablated versions:

- **Spatial-CLAP (conventional):** Baseline framework utilizing a spatial encoder trained on a SELD task to estimate a single central DoA of a trajectory, even for a moving source. To align with our experimental constraints, its encoders were frozen during training. We also evaluated w/ time series, in which the spatial encoder was trained without global pooling, thereby preserving frame-level DoA estimates. This variant was introduced to investigate whether simply retaining temporal information is sufficient for modeling moving sound sources.
- **Moving-CLAP (ours):** Proposed framework integrating a pre-trained spatial encoder with BEATs, an SBA-equipped text encoder, and the proposed auxiliary losses. We also evaluated four ablation variants: (i) **w/o SBA**, replacing SBA with simple averaging; (ii) **w/o MAB**, using concatenation and an MLP instead of the MAB; (iii) **w/o attn. pool.**, using mean pooling instead of attention-based pooling; (iv) **w/o auxiliary loss**, trained with only the CLIP loss.

3.4. Evaluation metrics

To evaluate the representational ability of content and spatial information, we conducted retrieval tasks, specifically audio-to-text retrieval (A2T) and text-to-audio retrieval (T2A), using Recall@1 (R@1) and Recall@5 (R@5) [1]. The retrieval results were evaluated across three distinct levels: (i) **monaural**, using the original AudioCaps test set to evaluate the model’s ability to extract content information, independent of spatial information; (ii) **1-src all**, using the single-source test set to evaluate the combined content-spatial representation by retrieving from a large candidate pool formed by all combinations of sound events and 25 spatial patterns; and (iii) **spatial pattern retrieval (1-src spatial and 2-src spatial)**, which extends the evaluation framework of Spatial-CLAP to isolate spatial discriminability among 25 and 180 spatial patterns, respectively.

Additionally, we report the **mean similarity**, defined as the average cosine similarity between positive audio–text pairs, to analyze the overall alignment in the embedding space. To quantify how effectively the embedding space is structured by spatial patterns, we calculated the **silhouette coefficient** [36]. This metric evaluates the cohesion of samples within the same spatial pattern versus their separation from different patterns. Coefficients were computed in the original 512-dimensional space for all samples in both audio and text modalities.

3.5. Result

Overall results. Table 3 shows the results. Moving-CLAP significantly outperforms the conventional Spatial-CLAP across all metrics.

Table 3: Results of retrieval performance (R@1 and R@5 [%]) and embedding space analysis (Mean sim. and Silhouette)

Framework	Monaural		1-src all				1-src spatial		2-src spatial				Mean	Silhouette	
	R@1		R@1		R@5		R@1		R@1		R@5		sim.	Audio	Text
	A2T	T2A	A2T	T2A	A2T	T2A	A2T	T2A	A2T	T2A	A2T	T2A	(↑)	(↑)	(↑)
(Random)	(0.10)	(0.10)	(0.004)	(0.004)	(0.02)	(0.02)	(4.00)	(4.00)	(0.56)	(0.56)	(2.78)	(2.78)	(0.00)	(0.00)	(0.00)
Spatial-CLAP (conventional)	15.0	16.2	2.00	2.16	7.91	8.14	9.36	7.60	1.18	1.15	5.94	5.45	0.78	-0.02	-0.01
w/ time series	10.3	10.3	14.4	14.0	39.5	39.1	71.8	66.4	23.9	22.4	65.3	62.8	0.89	0.05	0.06
Moving-CLAP (ours)	23.5	21.5	27.4	24.8	60.5	58.6	98.9	99.0	44.5	44.3	85.8	85.9	0.70	0.29	0.30
w/o SBA	20.6	18.8	21.1	21.5	52.4	53.6	95.0	88.9	33.9	30.3	74.2	71.8	0.66	0.26	0.18
w/o MAB	19.3	20.2	27.8	25.0	62.0	59.4	98.8	98.9	44.1	44.0	84.1	83.6	0.67	0.29	0.27
w/o attn. pool.	20.1	19.4	26.4	24.1	60.4	58.6	99.0	99.0	44.4	44.6	86.0	85.7	0.70	0.29	0.30
w/o auxiliary loss	25.8	22.5	27.2	24.4	60.3	57.2	96.7	92.5	31.6	28.2	74.5	70.8	0.73	0.14	0.16

Table 4: Performance comparison and embedding space analysis across various settings for the loss weights (λ_{sw} , λ_{sp}).

Param.	Monaural		1-src sp.		2-src sp.		Mean sim.	Silhouette	
	A2T	T2A	A2T	T2A	A2T	T2A		Audio	Text
Random	0.10	0.10	4.00	4.00	0.56	0.56	0.00	0.00	0.00
(0.0, 0.0)	25.8	22.5	96.7	92.5	31.6	28.2	0.73	0.14	0.16
(0.3, 0.0)	23.6	25.0	98.8	98.8	42.4	42.6	0.69	0.21	0.33
(0.3, 0.01)	23.5	21.5	98.9	99.0	44.5	44.3	0.70	0.29	0.30
(0.3, 0.1)	21.8	19.1	98.7	98.9	43.7	42.2	0.54	0.46	0.23
(0.0, 0.01)	24.8	21.9	98.0	96.5	39.1	33.1	0.71	0.25	0.17
(0.3, 0.01)	23.5	21.5	98.9	99.0	44.5	44.3	0.70	0.29	0.30
(0.6, 0.01)	24.2	20.4	98.8	99.0	44.5	44.7	0.70	0.29	0.31

Specifically, it achieves an R@1 score of nearly 99% for 1-src spatial. The w/ time series achieves better performance than Spatial-CLAP, indicating that finer temporal resolution is crucial for capturing moving sources.

An analysis of the failed retrieval samples in 1-src all reveals that more than 98% of the errors stem from the misclassification of content information rather than spatial information. These results suggest that capturing content information in complex audio signals remains a bottleneck.

Ablation study. Without SBA, both the R@1 score for 1-src spatial and the silhouette coefficient in the text embedding space decrease. Moreover, removing auxiliary losses also degrades both retrieval performance and the silhouette coefficient. These results indicate that SBA, swap loss, and spatial contrastive loss are essential for aligning the temporal changes of moving sound sources with their spatial information in captions and effectively structuring the embedding space.

Regarding feature fusion and pooling, simple concatenation (w/o MAB) or mean pooling (w/o attn. pool.) led to only a slight performance decrease across most metrics. These results suggest that the core spatial capability is primarily driven by the encoder architecture and auxiliary losses.

Parameter sensitivity. Table 4 shows the performance of Moving-CLAP with its various settings for the auxiliary loss weights. Increasing λ_{sp} leads to a higher silhouette coefficient in the audio embedding space, while increasing λ_{sw} raises the silhouette coefficient in the text embedding space. These trends indicate that each loss successfully structures its intended modality. While these losses are essential for spatial retrieval, increasing the loss weights (e.g., λ_{sp}) tends to degrade the R@1 score for monaural. This result demonstrates a fundamental trade-off that while auxiliary losses sharpen spatial clustering, excessive spatial regularization can harm the semantic alignment between audio and text.

Visualization of embeddings. We visualized the embedding space using UMAP [37]. After reduction to 50 dimensions via PCA [38], we projected 2,500 embeddings (25 patterns $\times$ 100 sources) for each modality. As shown in Fig. 2, w/ time series exhibits significant overlap

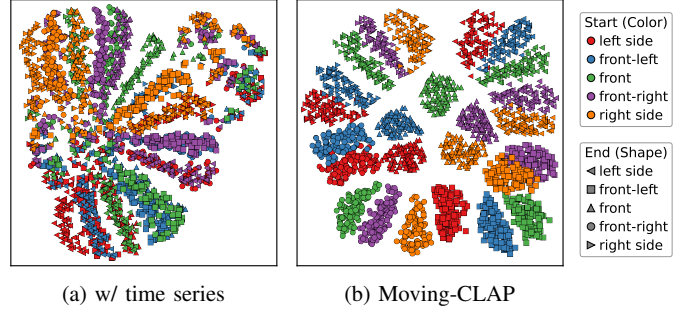


Fig. 2: UMAP visualization of the text embedding space. Colors and shapes denote the starting and ending positions of the sound sources, respectively.

across patterns, failing to capture the source motion. In contrast, Moving-CLAP forms an organized hierarchy where primary clusters, defined by end positions, contain five sub-clusters representing start positions.

4. LIMITATION

Although the proposed framework demonstrates higher performance than the conventional method on moving sound source retrieval, several limitations remain. First, all experiments were conducted using simulated spatial audio generated from monaural recordings. Therefore, further evaluation on real-world spatial recordings is necessary to verify the robustness of the proposed framework.

Second, the spatial captions were generated using predefined templates with a limited set of directional expressions. As a result, the current framework has not been validated for free-form natural language descriptions of moving sources, and extending it to more diverse spatial expressions remains an important direction for future work.

5. CONCLUSION

We proposed *Moving-CLAP*, a framework for the audio–text embedding of moving sound sources. Our results confirm that while the proposed auxiliary losses are indispensable for structuring the embedding space, balancing them with semantic alignment remains a key trade-off. Future work could address this trade-off by expanding the dataset to cover all spatial patterns for each audio source. In addition, we will evaluate the proposed framework on real-world data.

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